

Self-Study DA2 – Part II

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6. Instrument Validation

We restrict the sample to Region A ($N = 3,870$) and use the randomised **offer** as an instrument for **takeup**.

Random Assignment

Because the offer was assigned by lottery, it should be independent of pre-treatment characteristics. Table 1 reports mean differences and Welch t -tests for each covariate. Only **language_score** is significant at the unadjusted 5% level ($t = -2.72$, $p = 0.006$). With twelve simultaneous tests, the Bonferroni-corrected threshold is $0.05/12 \approx 0.0042$, and the imbalance is no longer significant. The joint F -test of all covariates on **offer** confirms balance: $F(12, 3857) = 1.27$, $p = 0.229$. The offer rate is 48.2%, indicating roughly balanced allocation. Figure 1 shows the standardised mean differences, all below the ± 0.1 threshold typically used as a balance benchmark.

Table 1: Covariate balance by offer status (Region A, $N = 3,870$)

Variable	Mean $Z=0$	Mean $Z=1$	Diff.	t	p
Age	35.09	35.01	-0.09	-0.31	0.759
Gender	0.52	0.51	-0.02	-1.00	0.319
Special trainings	0.66	0.65	-0.01	-0.37	0.711
Employment gaps	0.27	0.28	0.01	0.60	0.550
Baseline income	30479	30452	-26	-0.08	0.933
Unemp. duration	5.24	4.99	-0.25	-1.58	0.114
Language score	64.35	63.23	-1.12	-2.72	0.006
Immigrant	0.23	0.24	0.01	0.83	0.406
Disability	0.05	0.05	0.00	-0.11	0.912
Education: low	0.25	0.26	0.01	0.59	0.557
Education: medium	0.45	0.44	-0.02	-0.96	0.339
Education: tertiary	0.05	0.05	-0.01	-1.19	0.234
Joint F -test: $F(12, 3857) = 1.27$, $p = 0.229$.					

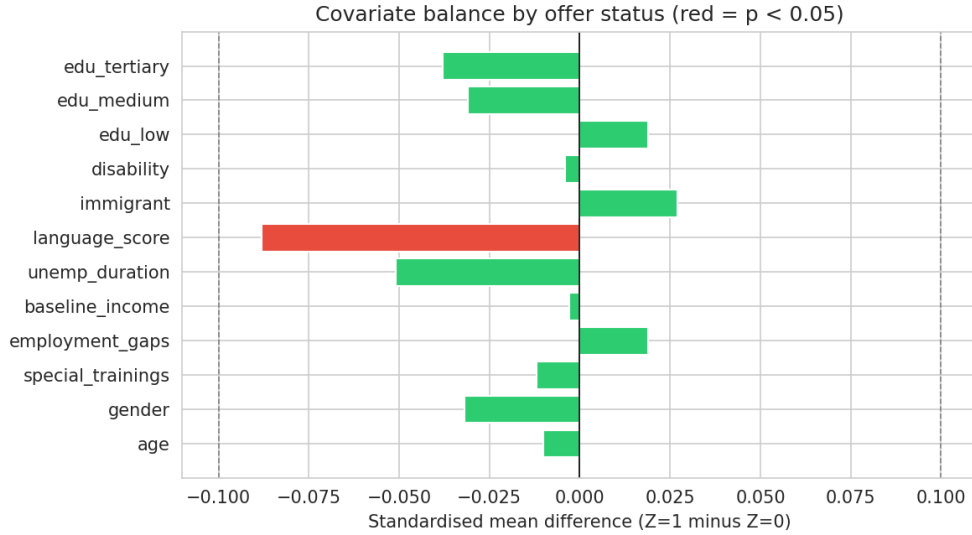


Figure 1: Standardised mean differences in covariates between offered ($Z = 1$) and non-offered ($Z = 0$) units. Dashed lines mark the ± 0.1 benchmark.

First Stage and Instrument Strength

Table 2 reports the first-stage OLS estimates and Figure 2 shows take-up rates by offer status. Being offered training raises take-up by **54.5 pp** ($SE = 0.011$, $t = 47.9$), stable with and without covariates, which confirms that the offer is unrelated to observables. The partial F -statistic for **offer** is **2,286**, far above the conventional rule-of-thumb $F > 10$, well above the Stock–Yogo critical value of 16.38.

Table 2: First-stage estimates (outcome: **takeup**)

	No covariates		With covariates	
	Coeff.	SE	Coeff.	SE
Intercept	0.012	0.008	−0.064	0.049
Offer	0.545	0.011	0.545	0.011
Partial F (offer): 2,286, $p < 0.001$. Compliance rate: 54.5%.				

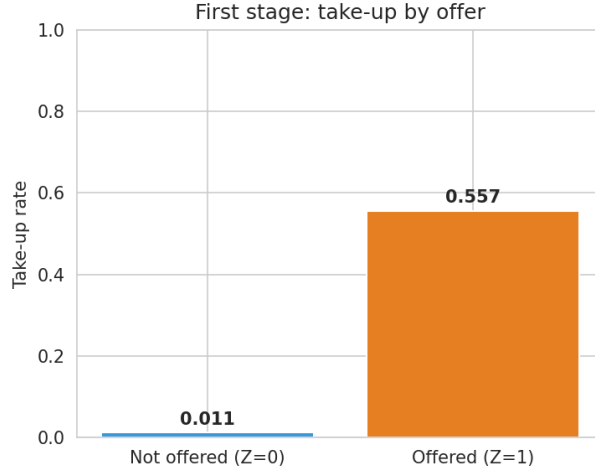


Figure 2: First-stage take-up rates by offer status. Take-up rises from 1.1% among non-offered units to 55.6% among offered units, yielding a compliance rate of 54.5%.

Relevance and Exogeneity

Relevance is satisfied: the offer raises take-up by 54.5 pp with an F -statistic of 2,286.

Independence and exclusion. Random assignment of Z makes it independent of unobservables, i.e. $Z \perp (Y^0, Y^1, D^0, D^1)$. The *exclusion restriction* additionally requires that Z enters Y only through D . Because the offer is purely a notification — no payment, no employer signal, no auxiliary counseling — the usual encouragement-effect channels that would violate exclusion are weak. The most plausible remaining violation is a direct effect on job-search effort, but for this to bias the LATE the channel would have to persist over the five-year horizon, which is unlikely.

Monotonicity is plausible but cannot be tested from the observed data; we assume it. Although a small share of always-takers exists (1.1% of non-offered units take up), there is no institutional mechanism that would make the offer *discourage* participation, so defiers can be ruled out. The LATE is therefore identified for the complier subpopulation.

7. Why not OLS

A simple OLS regression of **outcome** on **takeup** in Region A yields a coefficient of **2,848** (SE = 233, $p < 0.001$), suggesting that participants earn about 2,848 more after five years than non-participants. This estimate is not causally credible.

The core problem is self-selection on unobservables. In Region A, take-up is voluntary conditional on receiving the offer. Individuals who decide to participate are not a random draw: those who opt in are likely more motivated and have higher expected returns to training than those who decline. These same unobservable traits directly improve labour-market outcomes, independently of the training. Formally, $E[Y^0 | D = 1] \neq E[Y^0 | D = 0]$: the counterfactual outcome of participants, had they not trained, would still differ from the observed outcome of non-participants. OLS cannot separate the effect of training from these pre-existing differences, and adding observable covariates does not solve the bias. Importantly, OLS recovers neither the ATE nor the LATE under selection on unobservables: the bias contaminates both estimands.

The instrument solves this. The randomised offer provides exogenous variation in take-up that is unrelated to ability or motivation. Using **offer** as an instrument isolates only the variation in take-up driven by the lottery and removes the selection bias that biases OLS. Empirically, OLS gives 2,848 (SE 233) without controls and 2,762 (SE 180) with the full covariate set, both larger

than the IV LATE of 2,618. The gap closes only marginally with controls, which is itself consistent with selection on unobservables: covariate adjustment cannot remove what it does not observe.

8. IV Estimation

We estimate three causal parameters in Region A using **offer** as an instrument for **takeup**. All estimators are implemented from scratch: ITT and 2SLS use OLS via the normal equations, LATE uses the Wald formula, and standard errors for the Wald estimator are obtained by nonparametric bootstrap with $B = 500$ replications. ITT and 2SLS use analytical (homoskedastic) standard errors. Table 3 reports the results and Figure 3 visualises them with 95% confidence intervals.

Table 3: IV estimates of the effect of training in Region A ($N = 3,870$)

Estimator	Coefficient	SE	t	p
ITT (OLS Y on Z)	1,427.1	210.6	6.78	< 0.001
LATE (Wald)	2,618.4	375.7	6.97	< 0.001
2SLS (with covariates)	2,663.4	294.8	9.03	< 0.001

Wald SE: nonparametric bootstrap ($B = 500$).

Sanity check: 2SLS without covariates equals Wald exactly (2,618.4).

Homoskedastic SEs reported; HC1 robust SEs are 210 (ITT), 298 (2SLS), and 229 (naive OLS), suggesting that the homoskedasticity assumption is not driving inference.

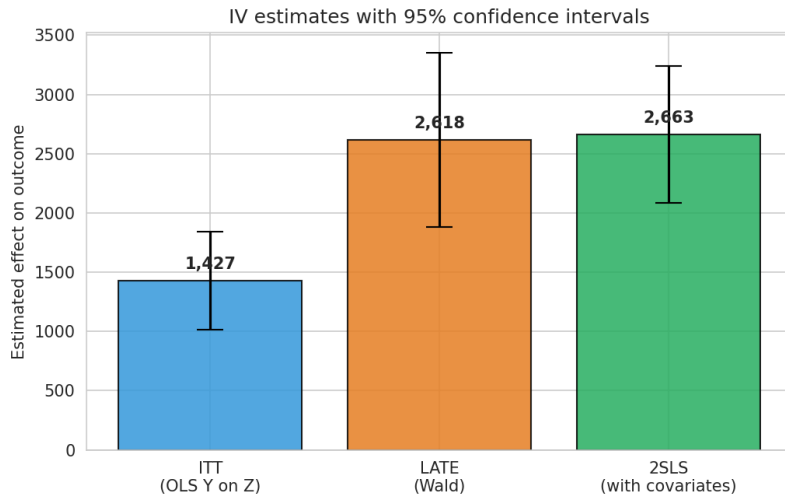


Figure 3: IV estimates with 95% confidence intervals. The LATE is roughly 1.83 times the ITT (the inverse of the compliance rate).

ITT. The intention-to-treat effect of 1,427 is the average gain in the five-year outcome from being *offered* the training, regardless of whether the individual actually participated. This is the policy-relevant quantity if the question is what the offer campaign actually delivers in the population. Since only 54.5% of those offered take up, the ITT is a diluted version of the participation effect.

LATE (Wald). Dividing the ITT by the first-stage compliance rate (0.545) yields the local average treatment effect of 2,618. This is the causal effect of *actually receiving* training, but only for the subpopulation of **compliers**: those who participate when offered and would not participate otherwise. It does not apply to never-takers or always-takers. The effect is statistically significant ($t = 6.97$, bootstrap SE = 376).

2SLS with covariates. Including covariates in 2SLS gives a similar point estimate of 2,663 with a tighter SE (295 vs. 376). Because the offer is randomly assigned, covariates are uncorrelated with

the instrument and adding them does not change the identification, only the precision. The smaller SE comes partly from the switch to analytical inference, not just from the covariates. The estimand is still the LATE; covariates only buy precision.

9. Comparing Identification Strategies

Table 4 and Figure 4 summarise the causal estimates from both parts. Part I uses Regions B and C ($N = 5,312$ after overlap trimming) under the Conditional Independence Assumption (CIA); Part II uses Region A ($N = 3,870$) with the randomised offer as an instrument.

Table 4: Causal estimates: Part I (CIA, B+C) vs Part II (IV, A)

Part	Estimator	Sample	Estimand	Coeff.	SE
I	Mean diff (raw)	B+C	ATE [†]	1,257	256
I	OLS (with cov.)	B+C	ATE	2,739	143
I	IPW	B+C	ATE	2,738	149
I	Doubly Robust	B+C	ATE	2,739	148
II	ITT	A	ITT	1,427	211
II	LATE (Wald)	A	LATE	2,618	376
II	2SLS (with cov.)	A	LATE	2,663	295

[†] Raw mean difference is biased without random assignment.

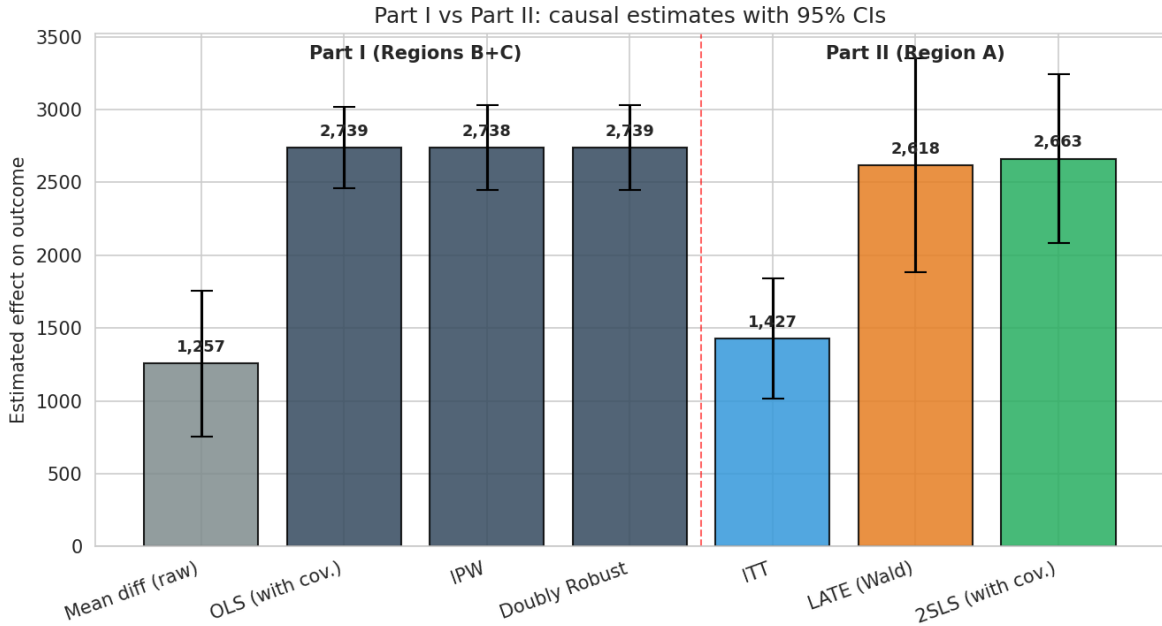


Figure 4: Causal estimates from Part I (Regions B+C, CIA-based) and Part II (Region A, IV-based) with 95% confidence intervals.

Validity of identification strategies. In Regions B and C, take-up is mandatory once assigned by caseworkers, so selection is plausibly on observables. The CIA is therefore credible and OLS, IPW, and DR all converge to roughly the same ATE ($\approx 2,740$). In Region A, take-up is voluntary, which makes selection on unobservables (motivation, expected returns) inevitable; the CIA fails. The randomised offer provides exogenous variation, so IV identifies the LATE for compliers ($\approx 2,620$).

ATE vs ITT vs LATE. The three estimands target different populations and questions. The ATE (Part I) is the average effect of training across all individuals in B+C. The ITT (Part II) is

the effect of being *offered* training, averaged over the whole population in A regardless of take-up; it is mechanically smaller than the LATE by the compliance rate ($1,427 \approx 0.545 \times 2,618$). The *LATE* is the effect of training itself, but only for **compliers**: those who take up if and only if offered.

ATE and LATE convergence. The Part I ATE ($\approx 2,740$) and the Part II LATE ($\approx 2,620$) are close, even though they target different populations. The closeness of the estimates is reassuring, but it should not be overinterpreted because the Part I ATE and Part II LATE refer to different regions and populations. This is consistent with a fairly homogeneous effect across subpopulations, although with these standard errors we cannot reject equality.

Policy implications. Three different estimands fit three different policy questions. For a voluntary offer-based program, the realised population-level effect is the ITT (1,427). The effect of training itself on those induced to participate is the LATE (2,620). If mandatory assignment by caseworkers is feasible (as in B/C), the relevant benchmark is the ATE (2,740). Mandatory assignment yields a larger effect on average but removes voluntary choice, which is itself a cost.